
A Tree-Based Framework for Student Performance Forecasting: Early Predictive Modelling and Individualized Behavioral Intervention

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Agenda

- ① Introduction to Learning Analytics
- ② Setting the Learning Environment
- ③ Forecasting Methodology
- ④ Behavior Adjustments
- ⑤ Discussion
- ⑥ Conclusions

A Brief Introduction to the Context

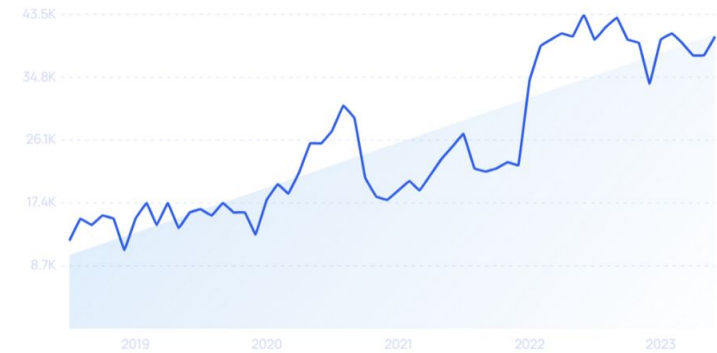


“The combined use of computer software and educational theory and practice to facilitate learning.”

An explosion in the volume of data available:

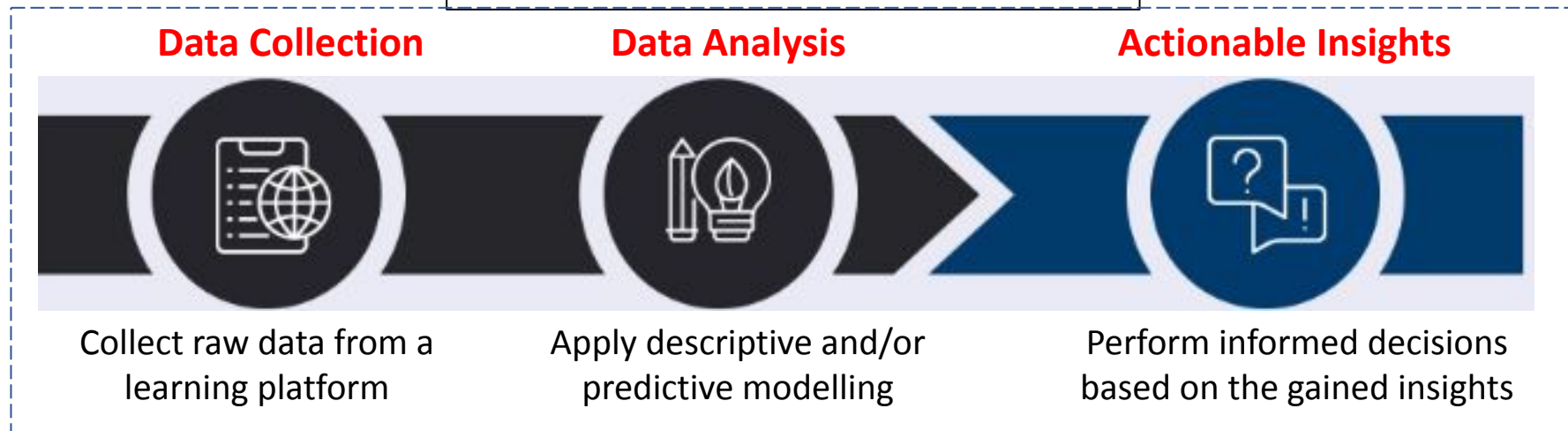
- Better understanding of the past
- More accurate future predictions

80% of college students say EdTech solutions helped them improve their grades



Searches for EdTech

Learning Analytics In a Nutshell



Learning Analytics: Benefits and Pitfalls

Reported Benefits of Learning Analytics:

- Reduce the dropout rate
- Assist in curricula design
- Create a more personalized learning environment
- Provide indications of students exam performance

The untold part of the story:

1. Accuracy over early predictions
Forecasting late leaves no time for learning adjustments
2. Teachers lack of confidence in complex predictive models
Limited faith in predictions they cannot interpret



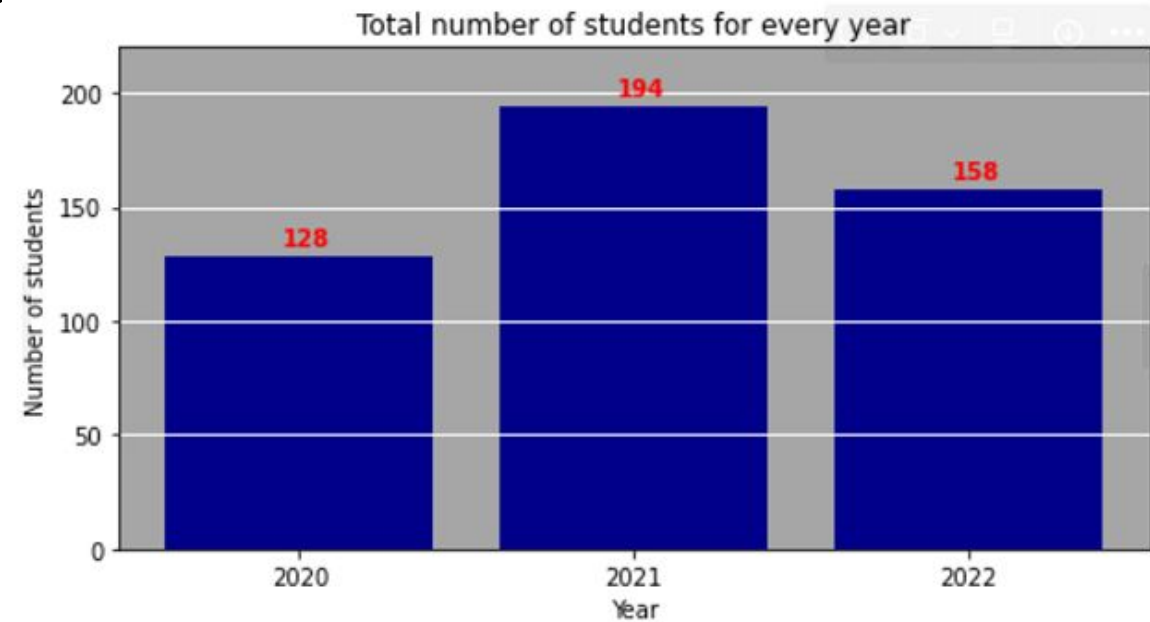
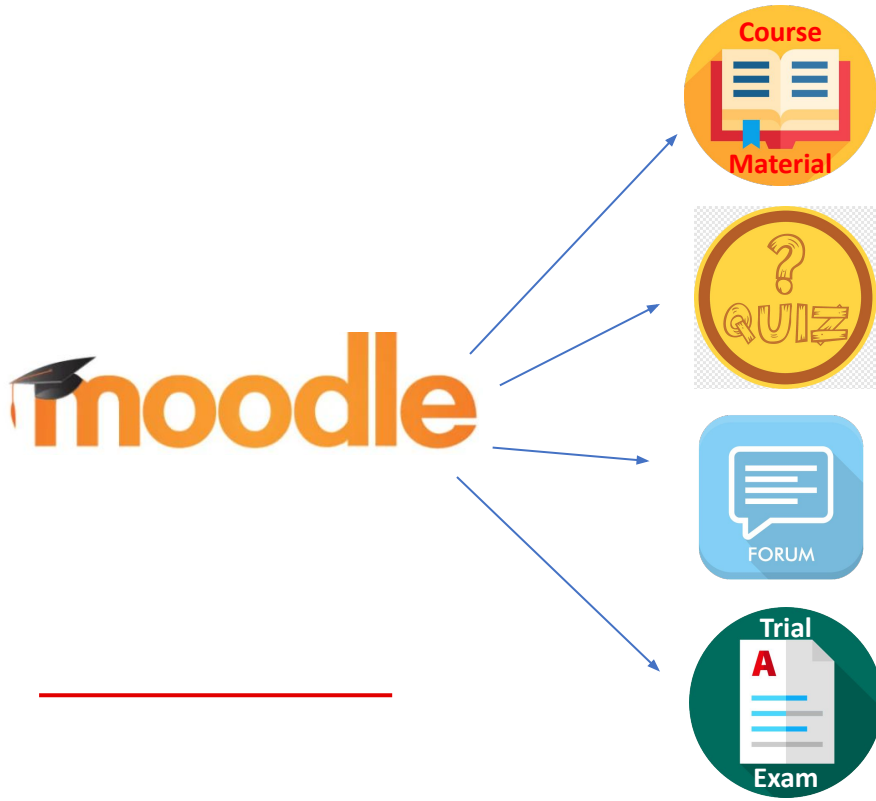
Vikos Canyon in north-western Greece

A Predictive Decision Support Framework

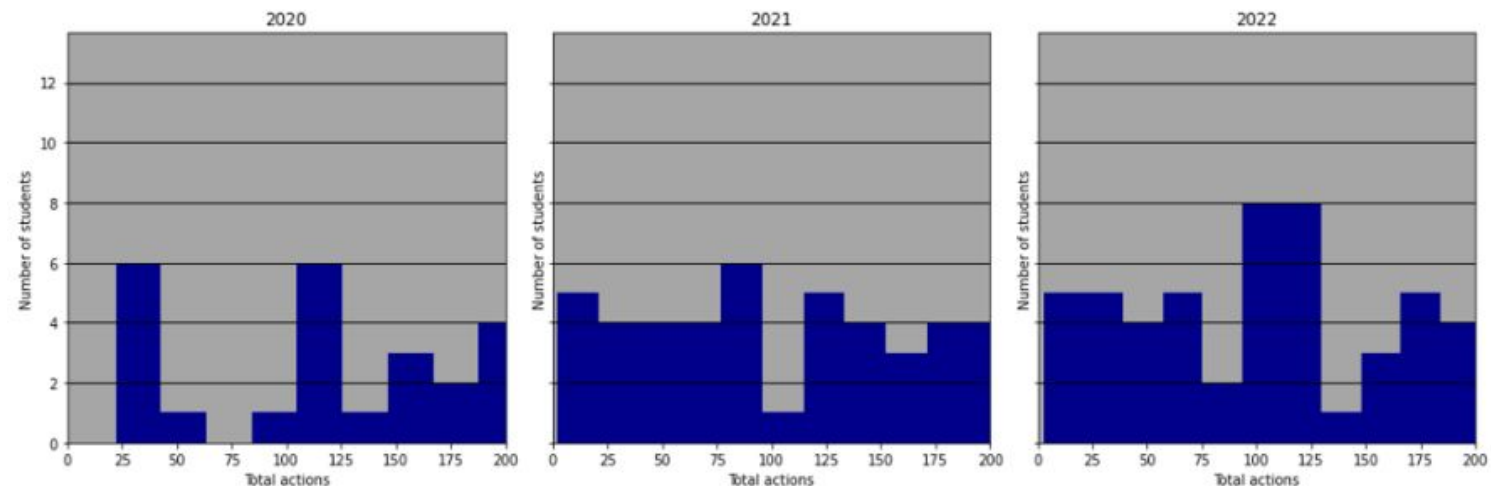
Setting up the Learning Environment

A single 1st year course.

- Data collected from 3 academic years
- Only the 1st semester is considered
- 2 years Remote Learning – 1 year Hybrid Learning



Potential issue with Data Shift?



An Interpretable Decision Support Framework:

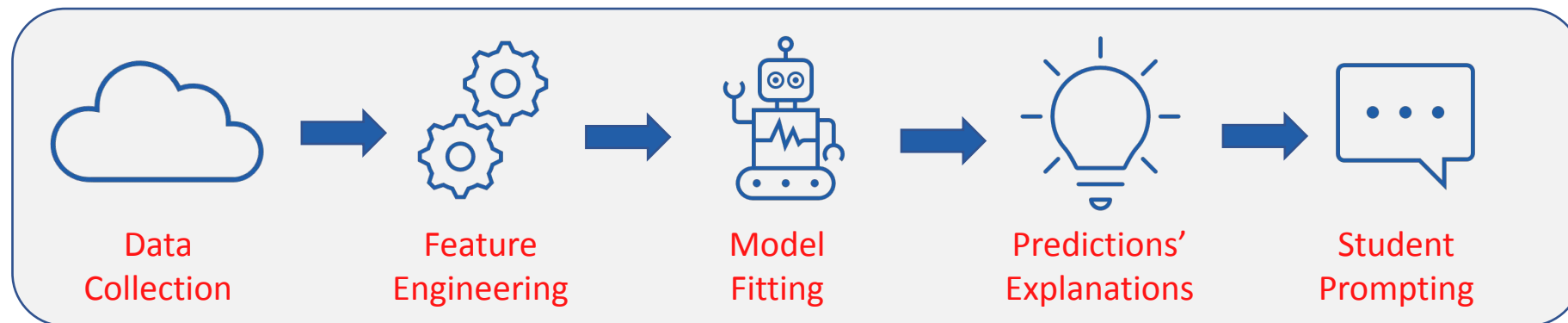
Using Predictive Modelling to Boost Students Exam Performance

The Objective of this Work: A Predictive Decision Support Framework 

Out aim is:

- Predicting Early 
- Predicting Accurately 
- Predicting Interpretably 

The Proposed Framework

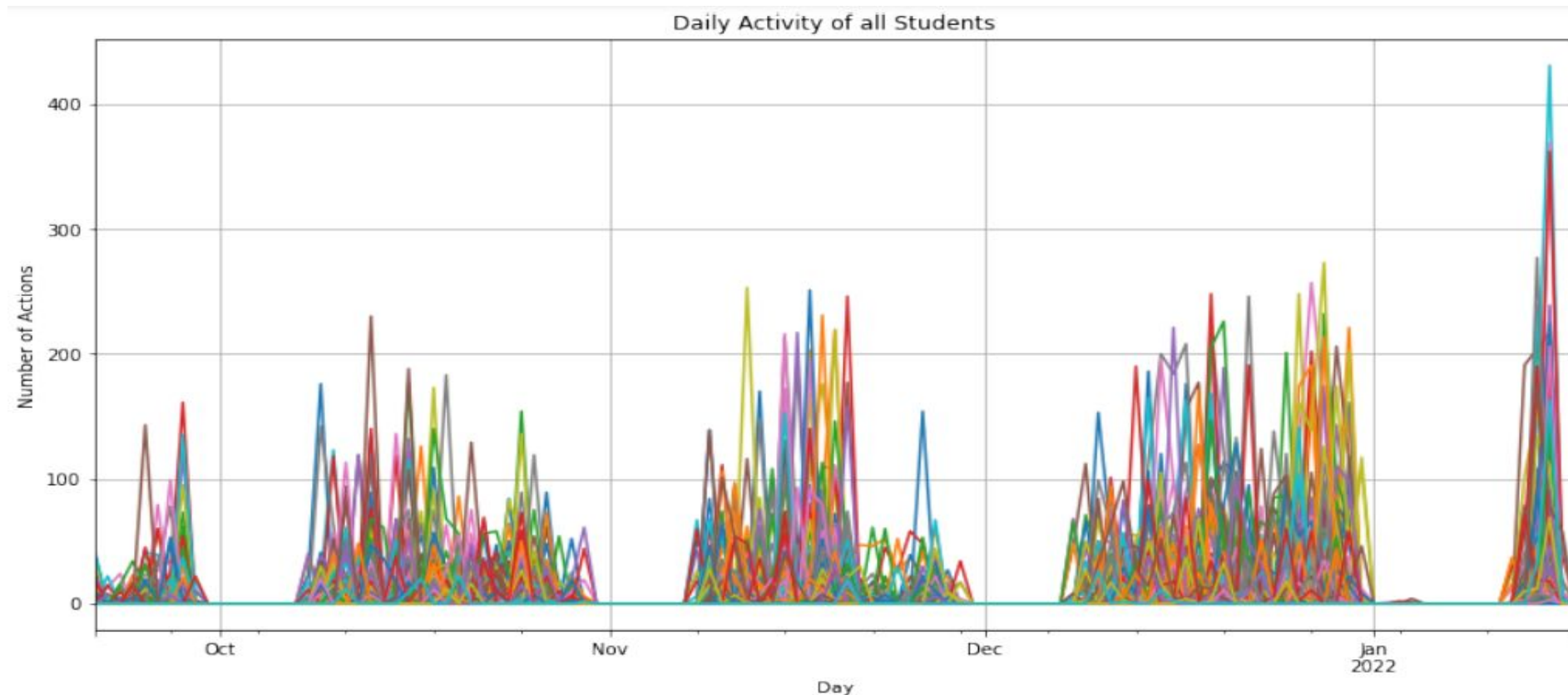


Forecasting Methodology:

1. Predicting Early

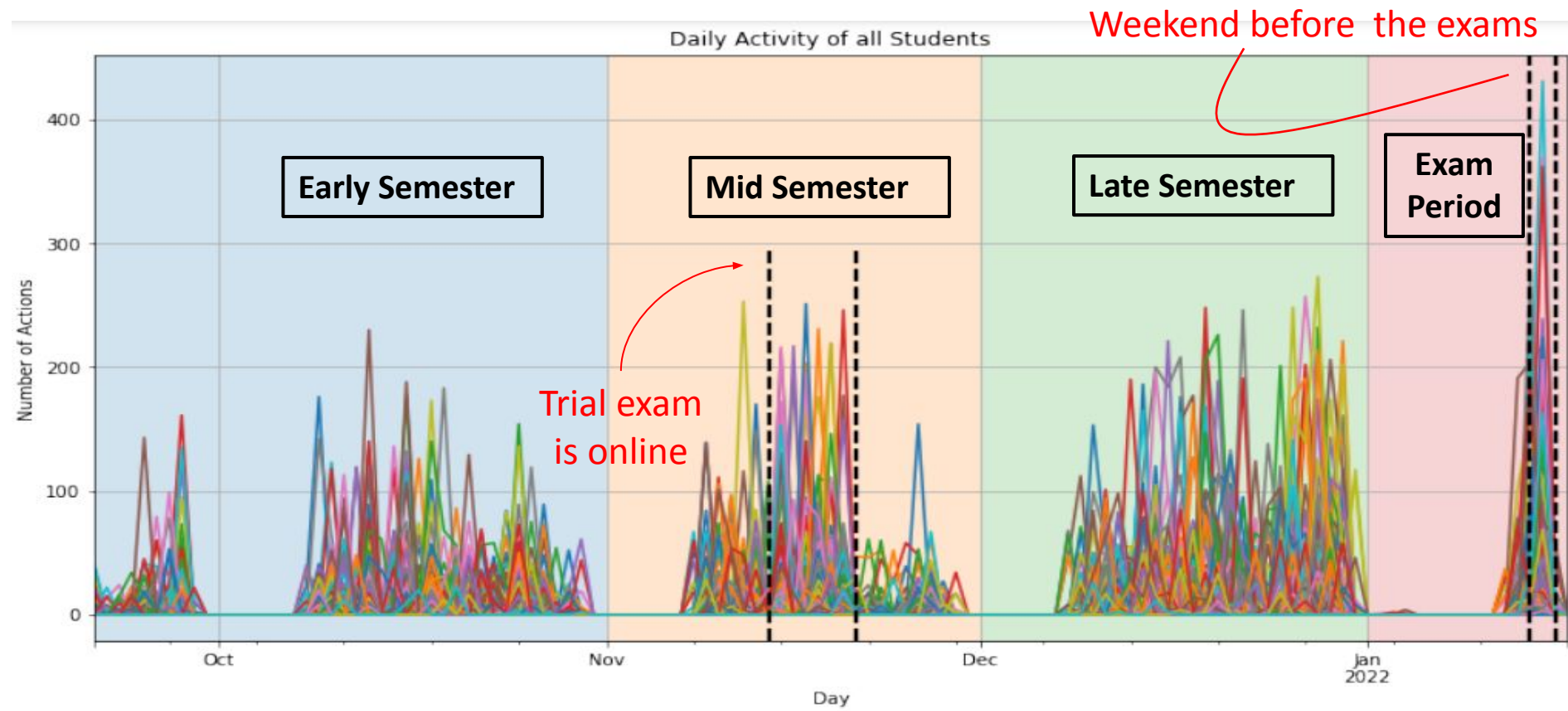
Forecasting Date Selection

- Leave Enough Time for Behavioral Adjustment



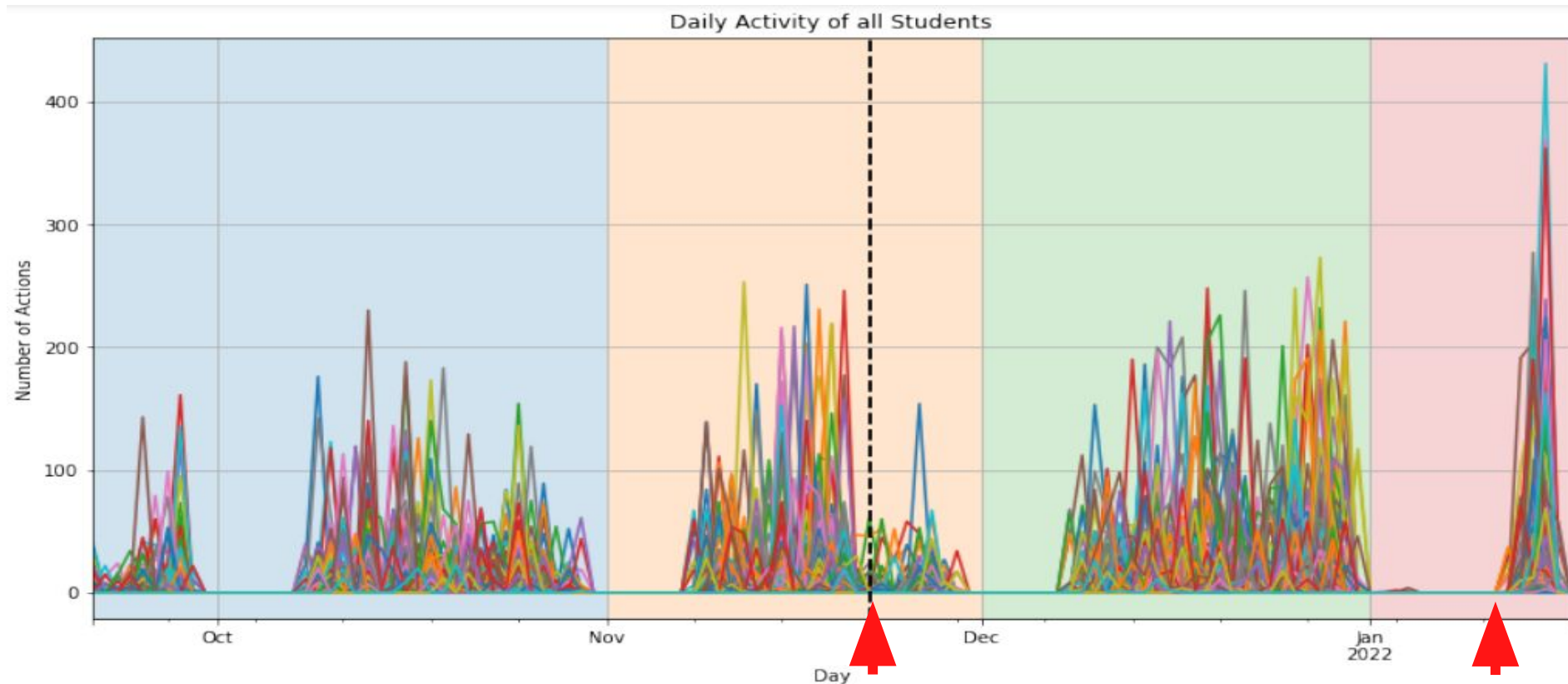
Forecasting Date Selection

- Leave Enough Time for Behavioral Adjustment



Forecasting Date Selection

- Leave Enough Time for Behavioral Adjustment



Our picked
Forecasting date

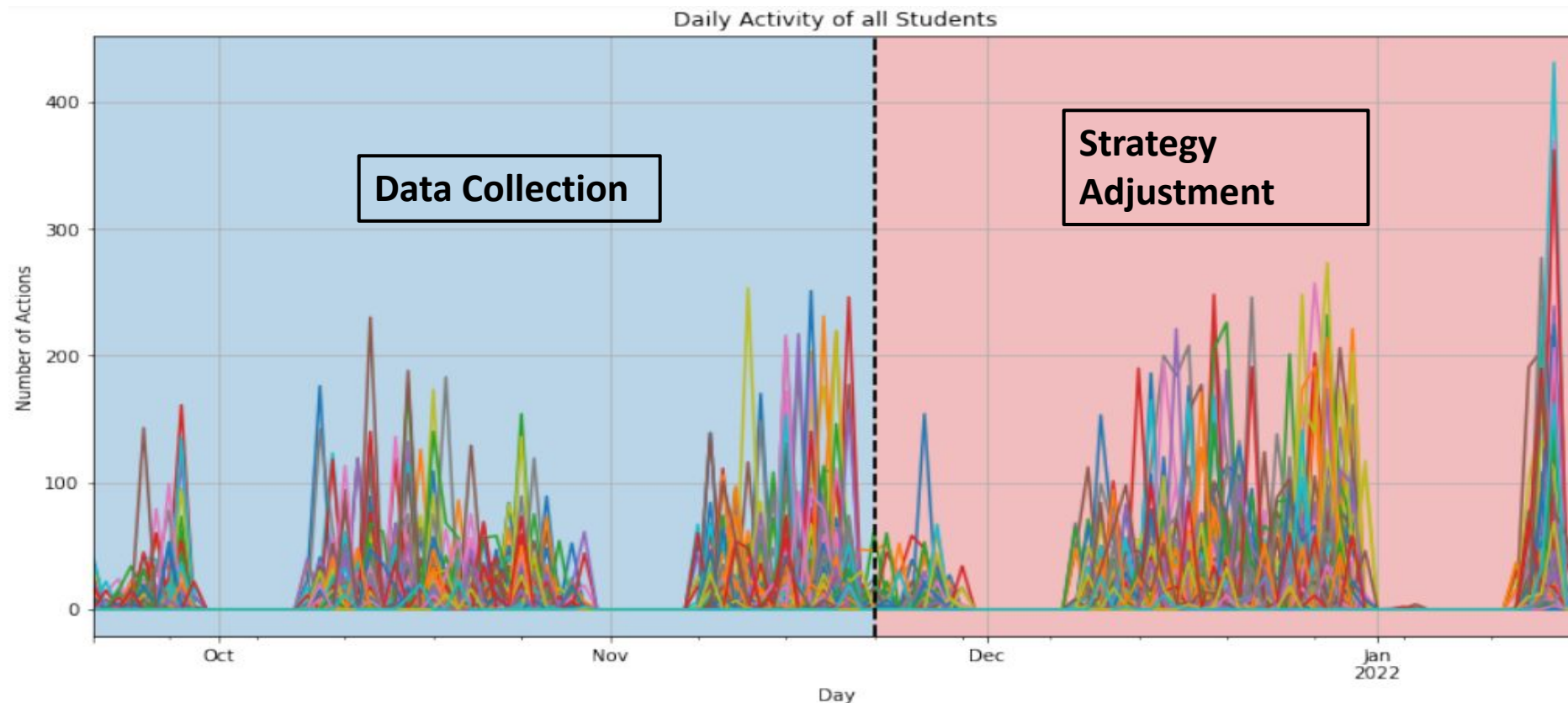
Most common
Forecasting date

Forecasting Date Selection

- Leave Enough Time for Behavioral Adjustment

Collect Activity data for 55 days

Leave 61 days for learning adjustments



Forecasting Methodology:

2. Predicting Accurately

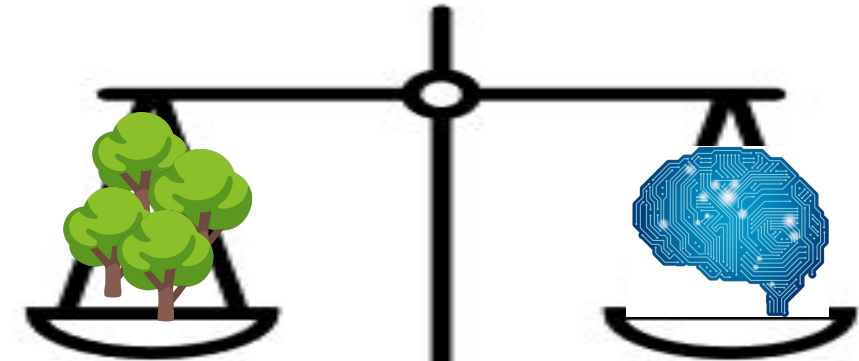
Model Selection: The thought process

Not entirely a forecasting task ➡ Statistical methods not applicable.
We Rely on Data-Driven Machine Learning Methods.

Two prediction paradigms, one big debate.

Which one to pick? 🧐

Conflicting literature findings



- Successful Feature Engineering
- Proper Hyperparameter tuning
- Skewed, longer tail feature distributions

- Correct Architecture Design
- Enough curated Data
- Regular and smooth feature distributions

Gradient Boosting Decision Trees: A good team of weak members

1. Different weak learners for diverse features
2. Informative feature space => more holistic view of the task
3. Requires detailed description of the problem

Neural Networks: A problem solver with unique pattern recognition skills

1. Multiple affine transformations
2. Finds meaningful representation of the feature space via deduction
3. Requires enough past experience and good architectural design

Model Selection: Final Model Selection



Feature Engineering:
We construct multiple and advanced features from raw data.



Model Selection

- We conduct an empirical investigation to identify the best model
- 5-fold cross validation on data of the first two years.

We pick CatBoost

Its not just more accurate!

- Handles missing values
- Allows the usage of multiple features
- Compatible with explainability techniques

Feature Set	Description
Regularity	Monitors Frequency of learning habits
Engagement	Monitors time spent on online course material
Performance	Monitors quiz and trial exam performance
Participation	Monitors participation in course related modules
Persistence	Monitors extent of task repetition
Student Profile	Personal characteristics and self-report learning traits

Bin students into 3 classes:

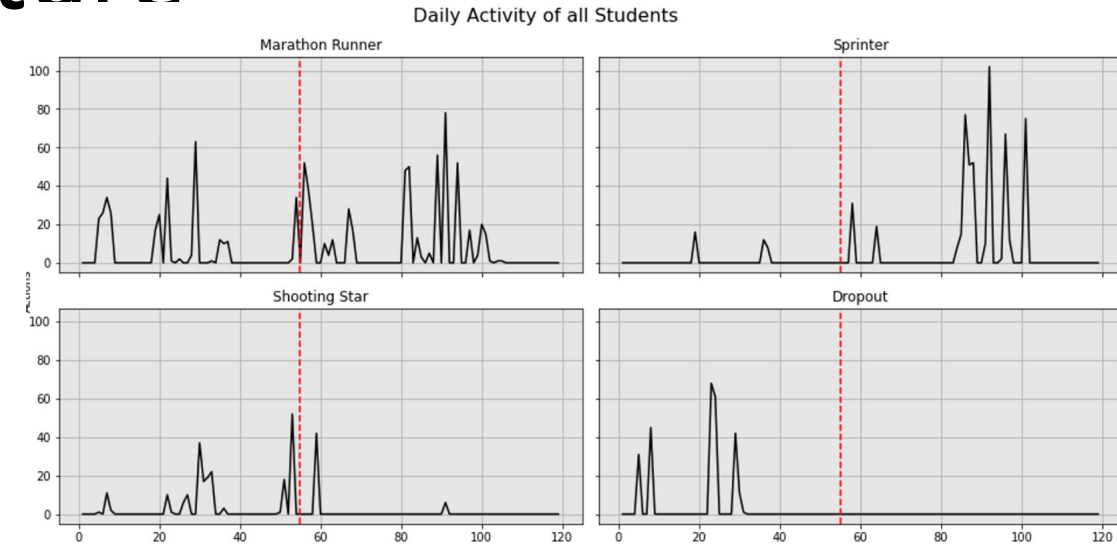
1. Grade < 8
2. Grade >8 & Grade < 13
3. Grade >13

Model	MSE	Recall	Accuracy
Logistic Regressio	-	0.38	0.4
XGBoost	10.83	0.68	0.53
CatBoost	10.45	0.71	0.55
LGBM	11.86	0.67	0.52
TabNet	14.58	0.65	0.42
FTTransformer	13.80	0.66	0.45

Embracing the Uncertain Future

How much faith can we put in a 1.5-month forecast of student performance?

- “Late Sprinters”, “Marathon Runners” and “Sugared-Up Sprinters”
- Unpredictable events: (eg break-ups, motivation adjustments)
- Pure luck?



Accuracy is just an illusion.

Enter Conformal Predictions (aka Empirical Methods)

Step 1: Estimate the conformal score: $\epsilon = f(y_{val} - \hat{y}_{val})$ on a validation set

Step 2: Estimate all quantiles q_a for $a \in (1,99)$ on ϵ

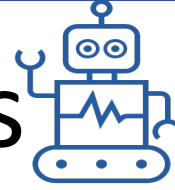
Step 3: For every prediction find q_a such as $\hat{y}_i \pm q_a \notin c_i$ with c_i the binned class of $\hat{y}_i$

Step 4: Interpret q_a as the confidence in the prediction of $\hat{y}_i$

Forecasting Methodology:

2. Predicting Interpretably

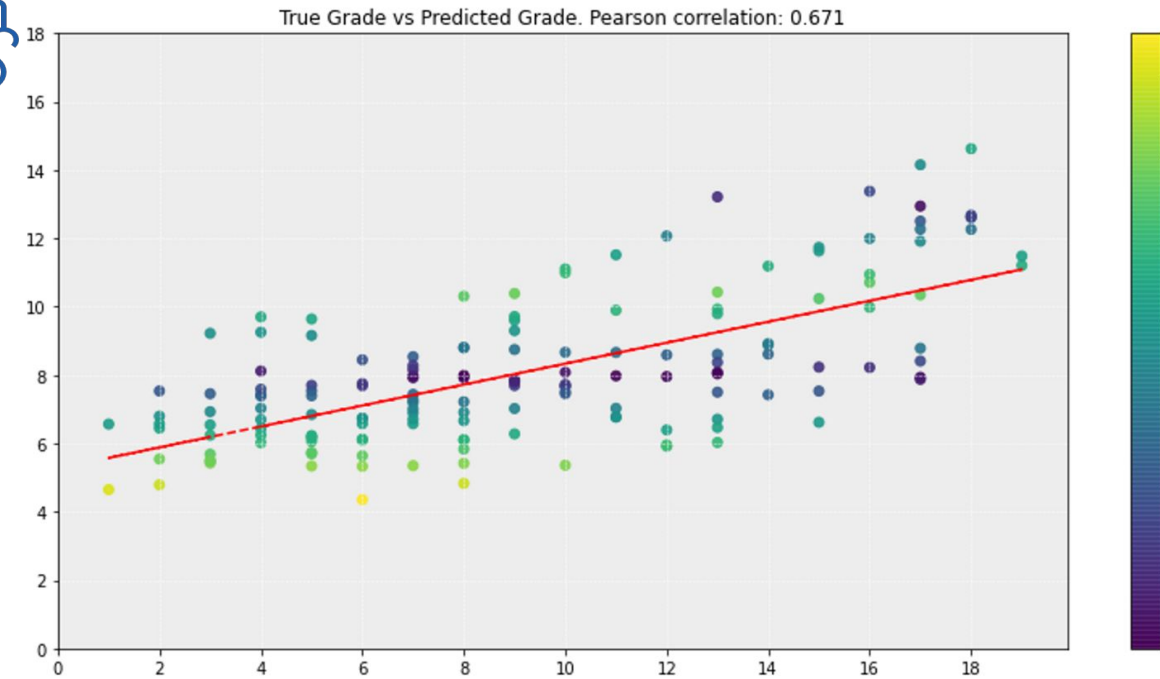
Explain your predictions



SHapley Additive exPlanations (SHAP) Values:

- Inspired by game theory
- Quantify the contribution of each feature to a single prediction
- Consider collaboration among features

A weighted sum of the marginal contribution of feature to a local prediction



 SHAP values measure feature contributions to predictions, not true grades

Why SHAP Values you ask.

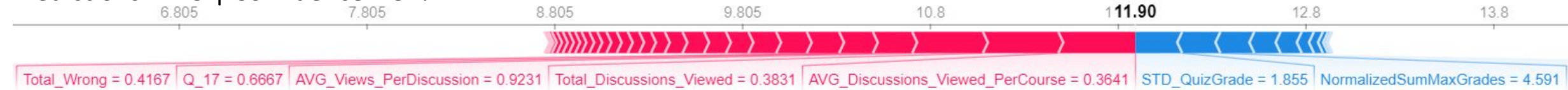


1. Identify which features contributed negatively to a student's predictions
2. Convert these features into areas of potential improvement for each student

Some Explanations...

Student A:

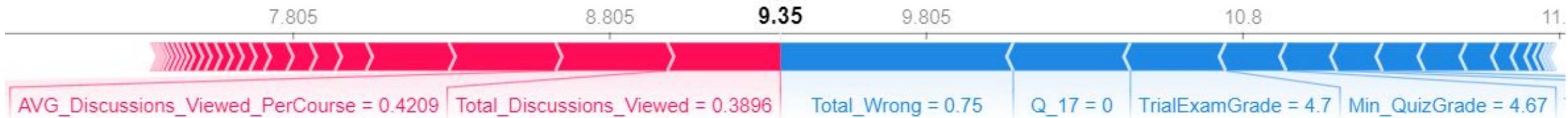
Predictions: 11.9 | Confidence = 51%



High Variance on Quiz Grades
Low Sum of Max Grades on Quizzes

Student B:

Predictions: 9.35 | Confidence = 82%



Bad Scores on Trial Exam
Low Minimum Score on a Quiz

Student C:

Predictions: 10.69 | Confidence = 67%



Low Forum Activity

Behavioral Adjustment

Turning Explanations into Prompts

Turn explanations into unique prompts:

Step 1: Match each feature with an appropriate prompt.

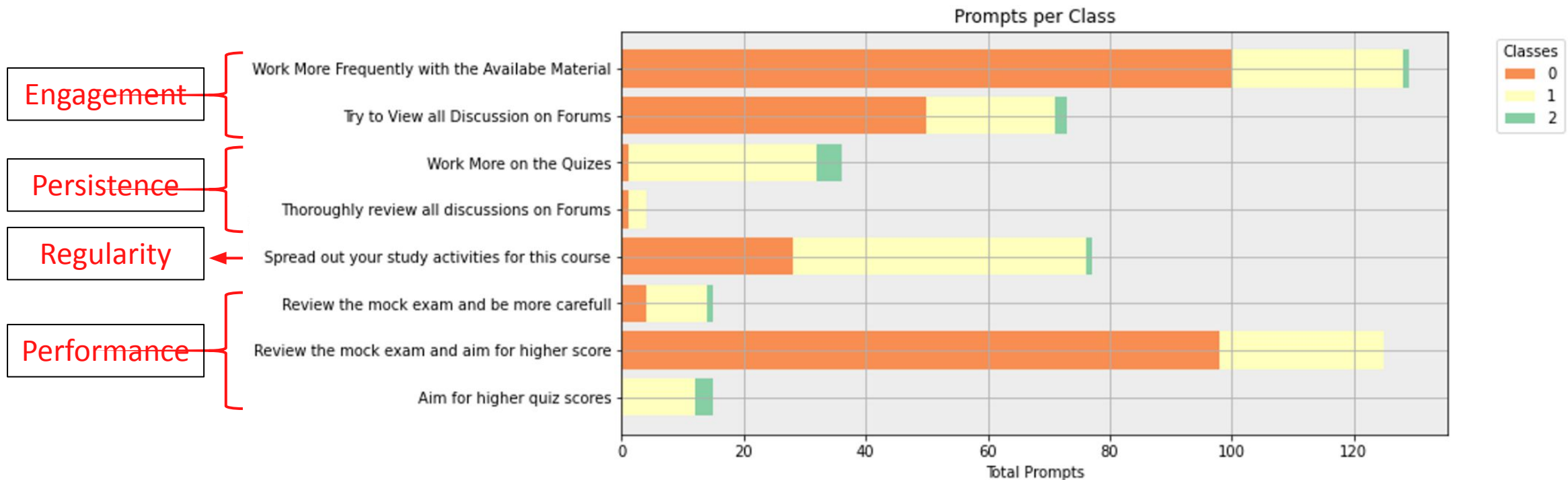
Step 2: Identify the 2 most negatively influencing features for every student

Step 3: Inform teacher for the prediction results and the prompt candidates

Step 4: Email students with the unique prompts.

Prompt Examples

Feature	Prompt
STD_Quiz_Grade	Work more on the quizzes
STD_Daily_Actions	Work more frequently on the material
AVG_Discussion_per_Course	Thoroughly review all discussions on Forums
TrialExamScore	Review the Mock Exam and aim for higher scores

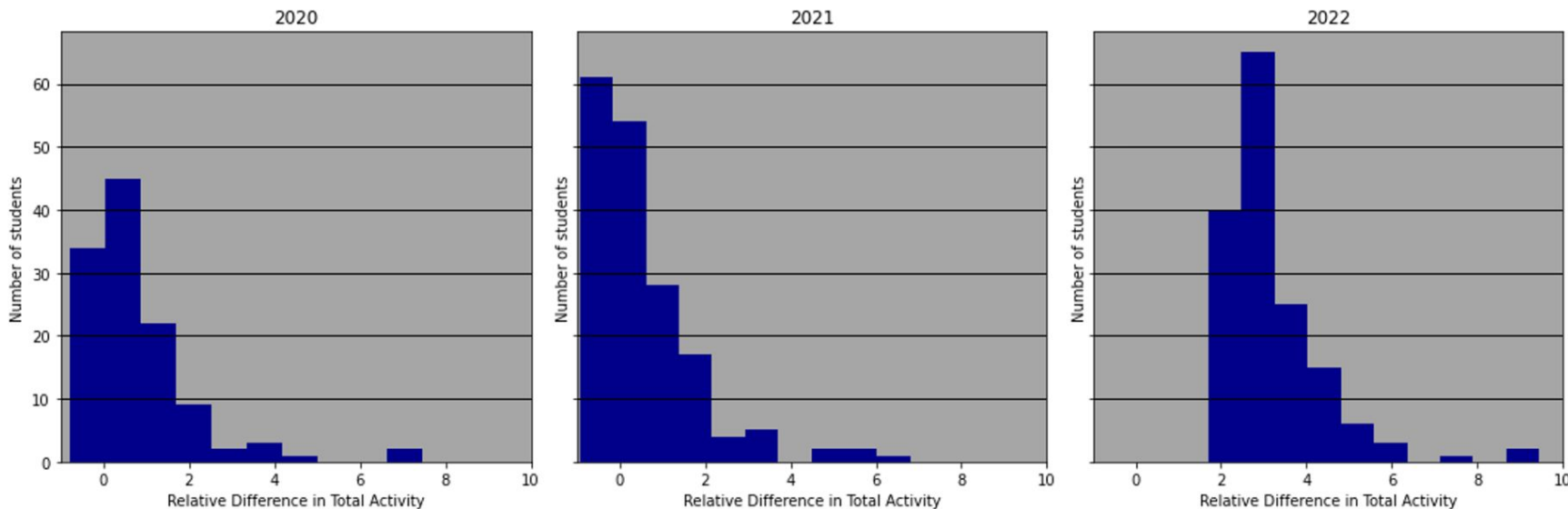


Did Prompting Worked? Some Initial Results

We define the **Relative Difference in Total Activity** as: $\log \frac{\frac{1}{n} \sum_{t=k+1}^T A_t - \frac{1}{n} \sum_{t=1}^k A_t}{\frac{1}{n} \sum_{t=1}^k A_t}$

In simple Terms:

How many more daily actions
(on average) a student made
after prompting



We conduct an ANOVA significance test

Comparisson	p-value	Significance
2022 vs 2020	2.66e-44	Yes
2022 vs 2021	3.51e-66	Yes
2022 vs 2020&2021	3.97e-78	Yes

Things to Discuss

1. We know prompting lead to learning adjustments.

Did it result in improvements in the exam performance?

We are investigating..

2. Can a fine tuned Large Language Model (LLM) become a great individual prompter?

Future Work: A personalized prompting assistant for every student

- Fine tune a LLM using Reinforcement Learning with Human Feedback (RLHF)
- Provide it with collected & predicted information for every student
- Collaborate with students.

To summarize:

We propose a **Predictive Decision Support Framework** for Learning Management Systems

We achieve:

- Early Predictions
- Accurate Predictions
- Transparent Predictions

Based on these predictions:

- We implement **specialized prompting**
- Initial results show **positive learning behavior adjustments**

In the Future:

We look to improve our prompting strategy using specialized AI agents for every student

The end 🥰

Questions?

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